

# Punctuated Semantic Chemistry as Homeostatic Inference Control

## Annealing, Alternating Phases, Causal Pruning, and Measurement

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### Abstract

This note summarizes a control principle for semantic graph algorithmic chemistry. The central distinction is:

Anneal when value is locally visible. Alternate when value is loop-level, delayed, and synergistic.

Semantic chemistry is specifically aimed at cases in which value is often loop-level and delayed: scientific paradigms, fictional worlds, self-models, relationship models, research programs, and other semantic autopoietic structures. Therefore the expected default is not pure continuous annealing, but punctuated, homeostatically regulated alternation: generate semantically rich, partially irresponsible loops; then test, compress, split, and prune them. The most important operational rule is:

Do not prune hard until the system has had enough time to form structures whose value is only visible after closure. Then prune ruthlessly. Not to make the system non-loopy, but to find which loops actually matter.

The main practical problem is measurement: how does a controller tell whether value is locally visible or distributed, delayed, and synergistic? This note proposes concrete diagnostics: paired replay, individual versus grouped ablation, maturation-lag tests, early-prune versus grace-period comparisons, ACS closure measures, effective evidence counts, compression curves, commutator defects, and novelty-quality trajectories. It also gives toy “just so” inference stories illustrating the distinction.

## 1 Background and vocabulary

A semantic chemistry state is a structured object

$$S_t = (G_t, R_t, C_t, A_t, L_t, M_t, E_t, W_t),$$

where, schematically:

- $G_t$  is a population of semantic graph molecules;
- $R_t$  is a soup of contextual graph rewrite rules;
- $C_t$  is a family of active context chambers;
- $A_t$  is an attention or resource allocation state;
- $L_t$  is a collection of event logs and provenance trails;
- $M_t$  is a set of causal modules and ACS macro-atoms;

- $E_t$  is a set of external tasks, evaluators, and reward models;
- $W_t$  is the worker-level scheduling state.

A typical semantic reaction rule has form

$$C \wedge G_1 \Rightarrow G_2,$$

where  $G_1$  and  $G_2$  are variable-bearing semantic graph patterns and  $C$  is a context graph. The rule may have a PLN-style intensional implication truth value, but the chemistry is not just a static database of implications. Rules, contexts, graphs, answer skeletons, story events, hypotheses, and macro-patterns interact as molecules, catalysts, and reaction products.

A semantic ACS is a self-supporting network of such transformations:

```
rules produce graph motifs
motifs activate more rules
rules and motifs close a loop
the loop generates answers, hypotheses, events, or self-model updates
```

A causal semantic ACS is stronger. It is not merely active or self-reinforcing. It matters under intervention:

```
ablating it changes downstream output
activating it changes downstream output
its utility is not fully reproduced by redundant paths
its effects are localized enough to be interpreted and gated
```

This is the distinction between a loop that merely keeps talking to itself and a loop that pays causal rent.

## 2 Two noncommuting operators

It is useful to idealize semantic chemistry control using two operators.

Let

$$G_T : S \mapsto S'$$

be a generative semantic chemistry operator at temperature  $T$ . High  $T$  means weak guidance, broader rule sampling, more analogy, more speculative bridging, more permissive evidence multi-counting, and less immediate causal discipline.

Let

$$P_\lambda : S \mapsto S'$$

be a causal pruning and compression operator at pressure  $\lambda$ . High  $\lambda$  means stronger ablation, duplicate-path taxation, causal-gate hardening, context splitting, compression, and removal of rules that have high correlational support but low intervention value.

In general,

$$P_\lambda G_T(S) \neq G_T P_\lambda(S).$$

Pruning before generation is not equivalent to pruning after generation. This noncommutativity is central. In autocatalytic search, a precursor may look useless before the loop closes and essential afterward. Early pruning can therefore delete the very catalysts needed for a useful semantic loop to become visible.

The control problem is to choose a schedule

$$(T_t, \lambda_t, \gamma_t, B_t, \rho_t, \sigma_t),$$

where  $T_t$  is temperature,  $\lambda_t$  is pruning pressure,  $\gamma_t$  is contextual or LLM guidance strength,  $B_t$  is budget,  $\rho_t$  is attention rent, and  $\sigma_t$  is a context-splitting threshold.

A simplified objective is

$$\max_{\{T_t, \lambda_t\}} \mathbb{E} [U(S_H) - c_{\text{cpu}}(S_{0:H}) - c_{\text{red}}(S_{0:H}) - c_{\text{drift}}(S_{0:H}) - c_{\text{double}}(S_{0:H})].$$

Here  $U$  is external utility: answer quality, hypothesis value, story coherence, prediction accuracy, self-model usefulness, or some mixture.

### 3 The core distinction

#### 3.1 Anneal when value is locally visible

A domain has locally visible value when useful inference fragments show their worth at short horizons and at small granularity. Let  $e$  be a rule, edge, path, module, or local graph motif. Define a short-horizon do-influence estimate

$$I_h(e) = \mathbb{E}[R_{t+h} \mid \text{do}(z_e = 1)] - \mathbb{E}[R_{t+h} \mid \text{do}(z_e = 0)],$$

where  $z_e$  indicates whether  $e$  is enabled, boosted, masked, or ablated.

Value is locally visible when:

1.  $I_h(e)$  is measurable with acceptable variance;
2. useful fragments have positive  $I_h(e)$  before large loops close;
3. grouped value is approximately predicted by local value;
4. causal estimates are stable enough to guide continuous gating;
5. early pruning does not destroy later discoveries.

Then a smooth schedule is appropriate:

$$T(t) \downarrow, \quad \lambda(t) \uparrow.$$

The system starts broad and becomes more selective. This is ordinary annealing. It works well for tasks like short diagnostic chains, local question answering, straightforward theorem-state transformations, and engineering debugging, where every good intermediate step has fairly immediate payoff.

#### 3.2 Alternate when value is loop-level, delayed, and synergistic

A domain has delayed synergistic value when useful structures are not visible at the level of individual rules. Let  $A$  be a candidate set of rules or graph motifs. Define a long-horizon grouped influence

$$I_H(A) = \mathbb{E}[R_{t+H} \mid \text{do}(z_A = 1)] - \mathbb{E}[R_{t+H} \mid \text{do}(z_A = 0)],$$

where  $H \gg h$  and  $z_A$  enables or disables the whole candidate module.

A loop-level delayed structure has the signature

$$I_h(e) \approx 0 \quad \text{for many } e \in A,$$

while

$$I_H(A) \gg 0.$$

Equivalently, the set has positive synergy:

$$\text{Syn}_H(A) = I_H(A) - \sum_{e \in A} I_H(e) > 0,$$

or, more operationally,

$$\text{Syn}_{h,H}(A) = I_H(A) - \sum_{e \in A} [I_h(e)]_+ > 0.$$

Here  $[x]_+ = \max(x, 0)$ .

This is the case for many semantic autopoietic structures: paradigms, fictional-world logics, self-models, relationship models, and research programs. Before the loop closes, many of the needed precursor motifs look like speculative junk. After closure, they become part of a coherent explanatory, narrative, or practical macro-atom.

The appropriate schedule is then not pure annealing but phase alternation:

$$S_{k+1} = P_{\lambda_k}^{m_k} G_{T_k}^{n_k}(S_k).$$

In words:

Run a permissive semantic chemistry long enough for loops to form. Detect candidate ACSs. Freeze, replay, ablate, compress, and prune. Compile the surviving loop as a macro-atom. Reheat around the surviving scaffold.

### 3.3 The likely default: punctuated annealing

The best practical regime is probably neither pure alternation nor pure annealing. It is punctuated annealing:

```
wild growth
-> stabilization
-> grouped causal testing
-> ruthless pruning
-> macro-atom compilation
-> context splitting if needed
-> reheating around the surviving scaffold
```

The amount of chaos gradually decreases across cycles, but within each cycle the system still permits a real generative phase before the pruning phase. This allows delayed loop-level value to become visible without letting entropy grow without bound.

## 4 The measurement problem

The central practical question is:

How can the controller tell whether value is locally visible or distributed, delayed, and synergistic?

This cannot be answered by introspection or by raw rule confidence. It requires measurements on the running inference ecology. The basic experimental tool is paired replay:

same snapshot  
same seed distribution  
same chamber budget  
same task or context  
condition A: rule/path/module/ACS enabled or boosted  
condition B: rule/path/module/ACS disabled, masked, cooled, or ablated  
measure reward, novelty, coherence, prediction error, compression, and drift

The replay target can be a single rule, a path, a context chamber, a candidate ACS, a semantic macro-atom, a worker, or an entire context update operator.

### 4.1 Individual versus grouped influence

For a candidate set  $A = \{e_1, \dots, e_n\}$ , estimate both individual and grouped influences.

Individual apparent value:

$$\text{AIV}_h(A) = \sum_{e \in A} [I_h(e)]_+.$$

Grouped intervention value:

$$\text{GIV}_H(A) = [I_H(A)]_+.$$

A simple local-visibility ratio is

$$\text{LVI}_{h,H}(A) = \frac{\text{AIV}_h(A)}{\text{GIV}_H(A) + \varepsilon}.$$

This number is not sufficient by itself, but it is useful together with signs, variance, and redundancy.

Typical interpretations:

- If  $\text{AIV}_h(A)$  is high, stable, and predictive of  $\text{GIV}_H(A)$ , value is locally visible.
- If  $\text{AIV}_h(A)$  is low but  $\text{GIV}_H(A)$  is high, value is delayed and synergistic.
- If  $\text{AIV}_h(A)$  is high but  $\text{GIV}_H(A)$  is low, apparent value is likely redundant or double-counted.

### 4.2 Synergy index

A normalized synergy index can be defined as

$$SI(A) = \frac{I_H(A) - \sum_{e \in A} I_H(e)}{|I_H(A)| + \sum_{e \in A} |I_H(e)| + \varepsilon}.$$

High positive  $SI(A)$  means the group does more than its members appear to do separately. This is a strong signal for alternating phases. Strong negative  $SI(A)$  means the members are redundant, mutually interfering, or over-counted. This is a signal for compression, path merging, or pruning.

### 4.3 Delay index

To detect delayed value, estimate influence across several horizons:

$$I_1(A), I_2(A), \dots, I_H(A).$$

A simple delay index is

$$\text{DI}(A) = \frac{\max_{\tau \in \{h+1, \dots, H\}} I_\tau(A) - I_h(A)}{|I_h(A)| + \varepsilon}.$$

High  $\text{DI}(A)$  means the module looks weak early but becomes useful later. A hard-pruning controller should back off when many promising candidate modules have high delay indices.

### 4.4 Early-prune versus grace-period test

A particularly practical measurement is the early-pruning fragility test. Run two matched traces:

Trace E: continuous causal pruning from the start

Trace G: grace period with weak pruning, then grouped pruning

Define

$$\text{EPF}(A) = R_{\text{grace}}(A) - R_{\text{early}}(A).$$

If  $\text{EPF}(A)$  is near zero or negative, early pruning did not hurt. Annealing is probably safe. If  $\text{EPF}(A)$  is strongly positive, early pruning killed a later useful structure. Alternating phases are indicated.

This test directly operationalizes the slogan:

Do not prune hard until the system has had enough time to form structures whose value is only visible after closure.

### 4.5 ACS closure and maturation

Delayed value is often linked to closure. A candidate ACS becomes more interesting when its members begin producing each other’s preconditions:

```
rule r1 produces motif m2
motif m2 activates rule r2
rule r2 produces motif m3
...
rule rk produces motif m1
```

Practical closure indicators include:

- the number and size of strongly connected components in the rule-motif graph;
- the fraction of active rules whose left-hand-side motifs are produced by other active rules;
- RAF-like closure: every rule has catalysts and reactants supplied by food or by the set itself;
- recurrence of the same loop across seeds, chambers, or tasks;
- increasing reward per CPU after the loop closes.

A controller should often use a grace period when candidate loops are approaching closure, even if local ablations remain inconclusive.

## 4.6 Reward density and temporal credit

Let  $r_t$  be incremental reward or evaluator score. In locally visible domains, reward is dense: useful rule firings are followed quickly by measurable gains. In delayed synergistic domains, reward is sparse or bursty: many apparently unhelpful steps are followed by a payoff after the loop closes.

A useful diagnostic is the distribution of credit assignment lags:

$$P(\tau \mid \text{event at time } t \text{ receives credit at } t + \tau).$$

Short, low-variance lags indicate local visibility. Long, high-variance lags indicate delayed distributed value. The latter supports phase alternation and longer replay windows.

## 4.7 Compression curve

After a generative phase, sort elements by estimated causal rent and prune to various retained sizes  $k$ . Measure

$$U(k) = \text{utility of the compressed subnetwork of size } k.$$

Interpretation:

- If  $U(k)$  degrades smoothly as  $k$  shrinks, value is relatively local and additive.
- If  $U(k)$  remains high until a loop core is removed and then drops sharply, there is a compact causal loop.
- If  $U(k)$  remains high after almost everything is removed, much of the wild phase was redundant; prune more aggressively next time.

## 4.8 Effective evidence count

For a conclusion  $c$ , suppose support paths  $p_i$  have weights  $w_i$  and pairwise provenance overlaps  $O_{ij} \in [0, 1]$ . A rough effective evidence count is

$$N_{\text{eff}}(c) = \frac{(\sum_i w_i)^2}{\sum_i w_i^2 + \sum_{i \neq j} O_{ij} w_i w_j + \varepsilon}.$$

If the raw path count is high but  $N_{\text{eff}}$  is low, the system is multi-counting the same evidence. This does not automatically mean delete the loop. It means: reduce confidence inflation, tax duplicate paths, merge near-equivalent rules, and use grouped causal tests.

## 4.9 Commutator defect

For two contexts  $c_1$  and  $c_2$ , let  $U_{c_1}$  and  $U_{c_2}$  be their update operators. Measure

$$\Delta(c_1, c_2; S) = D(U_{c_1}(U_{c_2}(S)), U_{c_2}(U_{c_1}(S))).$$

$D$  may combine answer-reward difference, graph distance, rule-weight distance, causal-gate distance, attention distance, ACS archive difference, and cost.

A small commutator defect means the contexts are compatible or well factored. A large defect means the system has context interference. The response may be:

split chambers  
 make context-specific rule copies  
 add an explicit bridge rule  
 tax overgeneral transfer rules  
 lower cross-context reuse  
 compile a guarded ACS macro-atom

High commutator defect often appears in self-model and relationship-model cases, where the order of emotional, social, and factual updates can substantially change the resulting model.

#### 4.10 Novelty-quality trajectory

Track novelty  $N_t$  and quality  $Q_t$  over time. Novelty can be estimated by nearest-neighbor distance from known motifs, known hypotheses, known story patterns, or known solution traces. Quality comes from task reward, evaluator judgement, prediction, coherence, or intervention value.

Typical signatures:

- Local-visible tasks:  $Q_t$  improves steadily as  $T$  decreases.
- Delayed-synergistic tasks:  $N_t$  may rise first,  $Q_t$  may wobble, and then  $Q_t$  jumps after a loop closes.
- Degenerate chaos:  $N_t$  rises while  $Q_t$  remains low and no causal ACSs survive ablation.
- Over-guided search:  $Q_t$  is safe but  $N_t$  is flat; reheat or weaken LLM control.

## 5 A practical classifier

The following table gives a measurement-oriented way to choose the control regime. Thresholds should be learned per domain; the table describes signs, not universal constants.

| Diagnostic          | Locally visible signature                      | Delayed synergistic signature                               |
|---------------------|------------------------------------------------|-------------------------------------------------------------|
| Individual ablation | Useful elements have stable positive $I_h(e)$  | Many members have weak or noisy $I_h(e)$                    |
| Grouped ablation    | $I_H(A)$ is predicted by individual influences | $I_H(A)$ is large despite weak individual influences        |
| Synergy index       | $SI(A)$ near zero or negative                  | $SI(A)$ strongly positive                                   |
| Delay index         | $DI(A)$ low                                    | $DI(A)$ high                                                |
| Early-prune test    | Early pruning does not reduce final reward     | Grace-period traces outperform early-pruned traces          |
| Reward density      | Reward follows useful events quickly           | Reward arrives after long chains or closure                 |
| ACS closure         | Closure not needed for value to appear         | Value appears around strongly connected or RAF-like closure |
| Compression curve   | Smooth degradation under pruning               | Sharp cliff when loop core is removed                       |
| Evidence count      | Independent support paths are clear            | Raw support high but independence unclear before pruning    |



|                      |                                                 |                                                                  |
|----------------------|-------------------------------------------------|------------------------------------------------------------------|
| Novelty-quality path | Quality improves steadily as guidance increases | Novelty rises before quality; quality jumps after loop formation |
| Recommended control  | Continuous annealing with on-going causal gates | Punctuated alternation with grace periods and grouped pruning    |

A compact decision rule is:

```

for each candidate module A:
    estimate individual influence I_h(e)
    estimate grouped influence I_H(A)
    estimate synergy, delay, early-prune fragility, closure, redundancy

if individual influence is reliable and group value is additive:
    use annealing
    decrease temperature gradually
    increase pruning continuously
elif group influence is high but individual influence is weak or delayed:
    use alternating phases
    give candidate loops a grace period
    prune only after closure or maturation tests
elif raw support is high but grouped influence is low:
    tax duplicate paths
    merge equivalent rules
    reduce confidence
    prune aggressively
else:
    hold/stabilize
    collect more replay evidence

```

## 6 Homeostatic regulation

A useful controller is homeostatic rather than fixed-periodic. It monitors state variables and adjusts temperature, pruning pressure, guidance, rent, and context splitting.

Potential setpoints include:

$$\rho(A_{\text{react}}) \approx 1 + \delta,$$

where  $A_{\text{react}}$  is an effective reaction/catalysis matrix. A spectral radius below one suggests a dead chemistry; far above one suggests runaway activation.

Other setpoints:

$$\begin{aligned}
Y_{\text{causal-ACS}} &\geq Y_{\text{min}}, \\
R_{\text{redundancy}} &\leq R_{\text{max}}, \\
Q_{\text{clarity}} &\geq Q_{\text{min}}, \\
\Delta(c_i, c_j) &\leq \Delta_{\text{max}}, \\
N_{\text{novelty}} &\text{ not too low and not ungrounded.}
\end{aligned}$$

One possible qualitative policy is:

|         |                                                              |
|---------|--------------------------------------------------------------|
| grow    | when the chemistry is dead or novelty is too low             |
| hold    | when candidate loops are forming but not mature              |
| prune   | when redundancy or confidence inflation dominates            |
| compile | when a loop survives grouped ablation                        |
| split   | when commutator defects are high                             |
| reheat  | when the compressed network is useful but no longer creative |

Thus oscillation is not necessarily failure. The desired behavior may be a limit cycle:

exploration -> loop closure -> causal testing -> pruning -> consolidation -> reheating

## 7 LLM guidance as a regulatable catalyst

An LLM or other contextually savvy model can help avoid trivial circularity, but it can also over-constrain creativity. Treat it as one catalyst, not the master controller.

A simple dominance diagnostic compares rule-selection distributions with and without LLM guidance:

$$D_{\text{LLM}} = D_{\text{KL}}(P_{\text{selected}}^{\text{with LLM}} \| P_{\text{selected}}^{\text{without LLM}}).$$

If  $D_{\text{LLM}}$  is high and novelty is low, LLM guidance is probably dominating the chemistry. Reduce guidance during generation. If coherence or safety collapses, increase guidance during pruning, explanation, and surface realization.

A useful division of labor is:

weak LLM guidance during early wild generation  
stronger LLM guidance during pruning, explanation, safety, and rendering

## 8 Just-so story 1: the blue-light memory gel

This toy scientific example illustrates delayed loop-level value.

### 8.1 Observations

01: Under blue light, the gel stiffens.  
02: After voltage pulses, the gel responds faster to later pulses.  
03: Damaged gel heals faster after oscillating electric fields.

A local geodesic controller gives three separate explanations:

01 -> photochemical crosslinking  
02 -> residual charge  
03 -> field-assisted polymer alignment

These are plausible but not yet a paradigm.

## 8.2 Wild semantic-chemistry phase

High-temperature rules fire:

```
blue light
-> radical-pair formation
-> local conductance change
-> ion-flow channeling
-> crosslink rearrangement
-> stiffness change
-> ion-flow pattern stabilization
-> memory of prior voltage pulse
-> faster future field response
-> repair/stiffening effects
```

Before the loop closes, several steps look weak or speculative. A local pruning operator might delete them. After closure, the loop becomes a macro-hypothesis:

```
The gel is an electro-mechanical memory material:
light perturbs conductance,
conductance guides ion flow,
ion flow rearranges crosslinks,
crosslinks stabilize future ion flow.
```

## 8.3 Measurement lesson

Individual ablations before closure give low value:

```
do(remove radical-pair bridge): little immediate effect
do(remove ion-flow stabilization): little immediate effect
do(remove crosslink-memory link): little immediate effect
```

Grouped ablation after closure gives high value:

```
do(remove electro-mechanical-memory-loop):
  01 is partly explained
  02 and 03 become much worse
  joint compression drops
  novel predictions disappear
```

This is a positive synergy and delay case. Alternation beats early pruning.

## 9 Just-so story 2: the census that moves houses

This toy fiction example illustrates world-rule autocatalysis.

### 9.1 World premise

In Anvar, a house exists at the address written in the royal census.  
Legal names, addresses, debts, and inheritance records are magically linked.  
If a person's legal name is shortened, the house loses rooms.  
If a family line is erased from the census, the district becomes unmappable fog.

A surface-level continuation might produce:

a corrupt official erases the hero's name  
the hero restores the record

This is coherent but generic.

## 9.2 Chemical loop

The semantic chemistry forms a native world-rule ACS:

```
tax debt
-> census penalty
-> legal name abbreviation
-> identity partial erasure
-> house loses room
-> neighboring addresses shift
-> census correction required
-> new tax assessment
-> more tax debt
```

A story event follows:

The Queen forgives famine debts for one night.  
Thousands of missing name-syllables return.  
Rooms reappear in houses whose addresses were computed while the rooms were gone.  
Streets lengthen, alleys fork, and neighborhoods push into one another.  
A prison appears inside a noble nursery because both inherited the same erased name.  
The Queen restored too much truth at once.

## 9.3 Measurement lesson

The event is good only if its causal support is native to the world rules. The pruning phase tests:

```
remove magic ink monopoly      -> event still works
remove ghosts                  -> event still works
remove name/address/debt loop  -> event collapses
remove inheritance duplication -> prison/nursery twist collapses
```

The retained causal core is:

```
legal record -> name identity -> address geography -> debt modification
-> mass restoration -> physical displacement
```

The loop is kept not because it is circular, but because it makes the world more productive, coherent, and plot-generating.

## 10 Just-so story 3: the self-model “I procrastinate because I am lazy”

This example illustrates a bad autopoietic loop: internally coherent, but negative under intervention.

## 10.1 Observations

01: I missed a deadline.  
02: I avoided opening the task document.  
03: I felt shame when thinking about the task.  
04: The task instructions were ambiguous.  
05: I did better on similar tasks when someone clarified the goal.

The wild phase permits several loops:

missed deadline  
-> evidence of laziness  
-> shame  
-> task avoidance  
-> more delay  
-> missed deadline

and also:

ambiguous task  
-> uncertainty about success criteria  
-> threat prediction  
-> avoidance of clarification  
-> task delay  
-> shame  
-> stronger threat prediction

The first loop has many apparent support paths, but many share the same root event. Raw evidence is high; effective independent evidence is low.

## 10.2 Causal pruning

Paired replay tests interventions:

do(boost LazyIdentity)  
do(suppress LazyIdentity)  
do(boost TaskClarification)  
do(suppress TaskClarification)  
do(boost SelfPunishment)  
do(suppress SelfPunishment)  
do(boost AmbiguityDetection)  
do(suppress AmbiguityDetection)

Suppose the results are:

LazyIdentity:  
    high narrative support  
    low predictive value  
    negative intervention value

SelfPunishment:

- increases shame
- increases avoidance
- lowers task completion

AmbiguityDetection:

- predicts avoidance episodes
- suggests useful intervention

TaskClarification:

- improves completion
- reduces shame

The system prunes the global lazy-identity loop and retains a more useful ACS:

- ambiguous task
  - > threat prediction
  - > avoidance of clarification
  - > delay
  - > shame
  - > higher threat prediction

The intervention rule becomes:

when task ambiguity is high:

- ask for success criteria early
- produce a first ugly draft
- prevent shame from becoming identity

### 10.3 Measurement lesson

This case warns that self-supporting does not mean good. The test is not whether the loop explains itself. The test is whether activating it improves prediction, action, repair, or self-regulation. The lazy-identity loop is semantically sticky but causally harmful.

## 11 Just-so story 4: a couple-level relationship model

This example illustrates multiparty self-models, evidence multi-counting, and commutator defects.

### 11.1 Events

E1: A sent a vulnerable message.  
E2: B replied six hours later with a short answer.  
E3: B had a difficult workday.  
E4: A did not know about B's workday.  
E5: A became cold at dinner.  
E6: B perceived A as punishing.  
E7: B withdrew.

A's chamber forms:

B slow reply

- > B is emotionally distant
- > B does not care about my vulnerability
- > I am unsafe if I depend on B
- > I should withdraw

B's chamber forms:

A cold at dinner

- > A is angry without telling me
- > I cannot win
- > I should shut down

Together they form a shared ACS:

ambiguous cue

- > negative attribution
- > defensive behavior
- > partner sees defensive behavior
- > partner makes negative attribution
- > partner defensive behavior
- > confirming cue

## 11.2 Causal pruning and bridge creation

A's model may count these as separate evidence:

late reply

short reply

quiet dinner

B looking at phone

B going to bed early

But several may be downstream of the same hidden event:

B's exhausting workday

Paired replay tests:

```
do(add bridge: B may be overloaded)
do(remove bridge: B may be overloaded)
do(add query action: ask what happened today)
do(remove query action)
do(boost attribution: B does not care)
do(suppress attribution)
```

Suppose the bridge and query action reduce conflict, while the attribution increases conflict and poorly predicts B's later explanation. The system keeps:

vulnerable bid

+ delayed response

+ missing context

+ threat attribution

+ defensive withdrawal

= conflict loop

It prunes the overconfident rule:

late reply ==> does not care

and replaces it with:

late reply ^ no work-context info ==> ambiguous cue

ambiguous cue ^ vulnerable-state ==> ask/check before attribution

late reply ^ known overload context ==> possible depletion, not necessarily rejection

### 11.3 Commutator lesson

The context updates may not commute:

U\_relationship\_threat(U\_work\_stress(S))

-> B was depleted; A was hurt; repair is possible.

U\_work\_stress(U\_relationship\_threat(S))

-> B always uses work as an excuse; threat model remains hot.

A large commutator defect says the model is not factored cleanly. The response is not only pruning, but bridge creation:

work stress can reduce responsiveness

reduced responsiveness can still hurt partner

hurt does not imply malicious intent

repair requires explicit context sharing

## 12 Just-so story 5: fire ecology

This toy scientific ecology example illustrates a loop whose members look weak until the consortium is considered as a whole.

### 12.1 Observations

01: Species A blooms after fires.

02: Species B appears near A but only in old burn zones.

03: Soil fungus F is abundant near both.

04: Seedlings of A survive poorly without B nearby.

A local explanation is:

fire -> nutrients -> A blooms

A changes soil -> B appears

fungus is incidental

The wild phase permits:

fire

-> mineral pulse

-> fungus F activation

-> fungus F supports A seedling



- > A root exudate supports fungus F
- > fungus F alters soil moisture
- > B germination improves
- > B leaf litter stabilizes fungus F
- > fungus F supports A seedling

The loop is:

A supports F  
 F supports A  
 F supports B  
 B supports F  
 fire activates F

## 12.2 Measurement lesson

Before closure:

F -> A survival            looks weak  
 B -> A survival            looks delayed  
 A -> F                    looks backwards

Grouped ablation reveals:

remove F rules:  
     initial fire bloom still partly explained  
     long-term persistence unexplained

remove B rules:  
     initial A bloom explained  
     old-burn-zone stability unexplained

remove fire rules:  
     loop lacks trigger

The retained macro-hypothesis is:

fire primes fungus  
 fungus supports A  
 A feeds fungus  
 fungus enables B  
 B stabilizes fungus

This is an alternating-phase case because local pruning would remove the weak mutualist bridges before the ecological loop becomes visible.

## 13 Just-so story 6: robot vacuum diagnosis

This contrast case illustrates when annealing is better.

## 13.1 Observations

01: Robot vacuum fails to dock.  
02: Battery is low.  
03: Dock LED is off.  
04: Robot docks successfully when placed on a different powered dock.

Rules:

dock LED off  
-> dock lacks power

dock lacks power  
-> robot cannot charge

robot cannot charge  
-> battery remains low

working alternate dock  
-> robot charging circuit likely okay

Every step has immediate causal payoff. There is no hidden high-order semantic loop that must mature before its value appears. Continuous annealing is appropriate:

high T:  
    consider robot fault, battery fault, dock fault, room-layout fault

medium T:  
    boost hypotheses explaining LED-off and alternate-dock success

low T:  
    converge on dock power problem

high pruning:  
    discard mapping confusion, battery memory effect, and fanciful hypotheses

## 13.2 Measurement lesson

Individual ablations are informative, reward is dense, grouped synergy is low, and early pruning does not kill later discovery. Annealing wins.

## 14 Good loops, bad loops, and causal rent

The crucial point is not to ban circularity. Both good and bad semantic systems are circular.

A bad loop says:

I believe X  
-> I interpret everything as evidence for X  
-> I act as if X  
-> my action creates more apparent evidence for X

Examples:

I am lazy.

My partner does not care.

The villain must be an evil scribe.

The fungus is incidental because plants are the visible actors.

A good semantic ACS says:

several weak motifs

-> mutually activate

-> form a higher-order explanatory object

-> survive ablation

-> compress many observations

-> generate useful interventions or story consequences

Examples:

electro-mechanical memory material

census-geography story world

ambiguity-threat-avoidance self-model

work-depletion / relationship-availability bridge

fire-triggered plant-fungus consortium

The operational difference is causal rent:

Does removing the loop hurt?

Does activating it help?

Does it explain independent observations?

Does it generate useful interventions or consequences?

Does it remain useful after redundant paths are compressed?

Does it stay inside its proper context?

## 15 Concrete implementation suggestions

### 15.1 Log enough to do replay

A system cannot measure delayed synergy without event logs. Each event should record:

rule fired

matched graph motifs

context chamber

upstream provenance

attention values

candidate ACS membership

reward-relevant outputs

random seed or sampling trace information

CPU and budget cost

This does not require perfect global proof trails. It requires enough provenance for local ablation, grouped ablation, and effective evidence counts.

## 15.2 Maintain candidate-loop watchlists

During high-temperature generation, do not immediately judge every fragment by local marginal value. Maintain watchlists of candidates that satisfy weaker signals:

part of an emerging strongly connected component  
part of a recurring cross-seed motif  
part of a high-novelty bridge  
part of a context-native world-rule interaction  
part of a module whose grouped value has not yet been tested

Apply lower rent to these candidates until maturation tests are run. This is not sentimental protection; it is a measurement strategy. The system is buying time to determine whether the apparent junk is precursor material for a useful loop.

## 15.3 Use three pruning levels

Not all pruning should be deletion. Use levels:

1. Soft tax: lower confidence, attention, or firing probability.
2. Compression: merge duplicate paths, rename a loop as a macro-atom, or replace many rules with a guarded module.
3. Deletion: remove or deactivate elements that fail ablation and have no useful context.

Premature deletion is dangerous in delayed-synergistic domains. Soft tax and compression are safer early.

## 15.4 Separate correlational, semantic, and causal scores

Do not collapse all rule quality into one weight. Track at least:

PLN intensional strength: semantic plausibility  
confidence: amount and diversity of support  
do-influence: interventional impact  
clarity: directness and nonredundancy of causal contribution  
novelty yield: production of nontrivial bridges  
context stability: whether the rule transfers safely  
CPU cost: resource burden

A rule can be semantically plausible but causally useless. Another can be weak as a local implication but crucial as a bridge in a loop.

## 15.5 Gate by phase

During generation:

reward novelty, bridge formation, ACS closure, and semantic fertility  
allow temporary double-counting  
weaken LLM domination  
avoid hard pruning of watchlisted precursors

During pruning:

```
run paired replay
estimate individual and grouped do-influence
tax redundant paths
compute effective evidence counts
split contexts with high commutator defects
compile surviving loops into macro-atoms
```

During reheating:

```
seed the chamber with compiled macro-atoms
explore anomalies and boundary cases
weaken gates enough for new loops to form around the scaffold
```

## 16 Summary

The rigorous distinction is:

Anneal when value is locally visible. Alternate when value is loop-level, delayed, and synergistic.

Local visibility means that short-horizon individual interventions are reliable guides to final value. Delayed synergy means that individual components look weak before closure, while the grouped module has significant long-horizon influence after closure.

Semantic chemistry is aimed precisely at domains where meaning is often organized in self-supporting loops: scientific paradigms, fictional worlds, self-models, multiparty relationship models, research programs, and other autopoietic semantic structures. Therefore punctuated, homeostatically regulated alternation should be the default. Smooth annealing is a special case for smoother, denser-feedback tasks.

The most important operational rule is:

Do not prune hard until the system has had enough time to form structures whose value is only visible after closure. Then prune ruthlessly. Not to make the system non-loopy, but to find which loops actually matter.

The practical heart of the matter is measurement. A controller should monitor individual versus grouped do-influence, synergy, delay, early-prune fragility, ACS closure, reward density, compression curves, effective evidence count, commutator defects, and novelty-quality trajectories. These measurements allow the system to decide whether to anneal, alternate, hold, prune, split, compile, or reheat.

## References

- [1] Ben Goertzel and GPT-5.5 Pro. Semantic Chemistry: Algorithmic Chemistry on Semantic Graphs as a Creative Inference Control Strategy. Draft manuscript, May 7, 2026.